Objective:

Autonomic nervous system (ANS)
Discuss the general organization and activation of ANS
Discuss structure and functions of sympathetic, parasympathetic nervous system and adrenal medulla
Compare the divisions of the ANS regarding origin of preganglionic fibers, location of ganglia and neurotransmitter substances
Discuss the value of adrenal medullae in the function of the sympathetic nervous system.

Physiology Practice Block: ***Autonomic Nervous System***

Q1. The preganglionic neurons of the sympathetic nervous system originate exclusively within the thoracolumbar segments (T1 to L2 or L3) of the spinal cord gray matter.
True / False
Q2. The parasympathetic nervous system is anatomically characterized as a craniosacral division, with preganglionic fibers emerging via Cranial Nerves III, VII, IX, and X, as well as sacral spinal segments S2 through S4.
True / False
Q3. In the sympathetic nervous system, the preganglionic nerve fibers are characteristically long and pass all the way to the effector organ before synapsing, whereas the postganglionic fibers are very short.
True / False
Q4. All preganglionic autonomic neurons—both sympathetic and parasympathetic—release acetylcholine as their primary neurotransmitter substance to excite nicotinic receptors on postganglionic neurons.
True / False
Q5. Parasympathetic postganglionic neurons are primarily adrenergic, releasing norepinephrine at their neuroeffector junctions to act on alpha and beta receptors.
True / False
Q6. The sympathetic postganglionic fibers that innervate most sweat glands and some blood vessels in skeletal muscle are exceptional because they release acetylcholine rather than norepinephrine.
True / False
Q7. Sympathetic chain ganglia (paravertebral ganglia) lie in a bilateral row along the vertebral column, allowing widespread, divergent sympathetic discharge when activated.
True / False
Q8. Terminal or intramural ganglia of the parasympathetic nervous system are located far away from the target organs, clustered primarily in the cervical region of the neck.
True / False
Q9. The adrenal medulla is embryologically and functionally a modified sympathetic ganglion, whose cells are innervated directly by preganglionic sympathetic fibers.
True / False
Q10. Upon sympathetic stimulation, the adrenal medulla secretes its hormones directly into the bloodstream, with the output consisting of approximately 80 percent norepinephrine and 20 percent epinephrine.
True / False
Q11. Circulating epinephrine and norepinephrine released by the adrenal medulla have the exact same short duration of action as the neurotransmitter substances released directly by sympathetic nerve terminals.

True / False

Q12. The presence of the adrenal medulla allows the sympathetic nervous system to exert a metabolic and physiological influence on cells and tissues that do not receive direct sympathetic nerve innervation.

True / False

Q13. Activation of the sympathetic nervous system causes a massive "fight-or-flight" response, which includes pupillary constriction (miosis), bronchoconstriction, and increased gastrointestinal motility.

True / False

Q14. Parasympathetic stimulation leads to conservation and restoration of bodily energy, manifesting as a decreased heart rate, pupillary constriction, and enhanced digestive secretions.

True / False

Q15. Muscarinic acetylcholine receptors are found on all postganglionic neurons within both the sympathetic and parasympathetic autonomic ganglia.

True / False

Q16. Alpha-1 adrenergic receptors are localized primarily on vascular smooth muscle, where their activation by norepinephrine triggers vasoconstriction.

True / False

Q17. Beta-1 adrenergic receptors are located predominantly in the myocardium, and their activation causes an increase in heart rate, conduction velocity, and myocardial contractility.

True / False

Q18. Activation of beta-2 adrenergic receptors results in the profound contraction of bronchial smooth muscle, significantly increasing airway resistance during exercise.

True / False

Q19. Dual innervation is a common feature of most visceral organs, where the sympathetic and parasympathetic divisions typically exert antagonistic (opposing) physiological effects to maintain homeostatic balance.

True / False

Q20. The autonomic nervous system operates entirely independently of any central nervous system input, receiving zero regulatory control from superior structures like the hypothalamus or brainstem reticular cores.

True / False

Physiology Practice Block: Autonomic Nervous System & Adrenal Medulla

Q1. A patient presents with an inability to sweat on the left side of the face alongside pupillary constriction. Which of the following parameters correctly identifies the anatomical origin of the preganglionic nerve fibers that coordinate sympathetic outflow to these cephalic structures?

a. The craniosacral parasympathetic nuclei of the lower midbrain

b. The dorsal root ganglia of the upper cervical spinal segments

c. The principal sensory nucleus of the trigeminal nerve pathway

d. The intermediolateral gray columns of the thoracolumbar spinal cord (T1 to L2 or L3)

e. The intramural ganglionic networks located within the skin tissues

Q2. The structural organization of the parasympathetic nervous system differs fundamentally from the sympathetic division. Which of the following choices correctly describes the location of the parasympathetic preganglionic cell bodies and their ganglia?

a. Thoracolumbar origin with paravertebral chain ganglia

b. Prevertebral ganglia receiving fibers from the lumbar cord segments
c. Sympathetic chain ganglia situated in a bilateral row along the spine
d. Collateral ganglia embedded within the internal elastic lamina of large arteries
e. Craniosacral origin with long preganglionic fibers that synapse in terminal or intramural ganglia located near or within the effector organs

Q3. When evaluating the length and myelination of autonomic nerve pathways, which of the following arrangements is characteristic of the sympathetic nervous system?

a. Long myelinated preganglionic fibers paired with short unmyelinated postganglionic fibers
b. Short unmyelinated preganglionic fibers paired with long myelinated postganglionic fibers
c. Myelinated fibers passing directly to skeletal muscle cells via monosynaptic paths
d. Short myelinated preganglionic fibers that synapse in paravertebral or prevertebral ganglia, giving rise to long unmyelinated postganglionic fibers
e. Unmyelinated fibers emerging from the cranium that bypass all peripheral ganglia

Q4. What specific neurotransmitter substance is secreted by all preganglionic autonomic neurons—both sympathetic and parasympathetic—to excite the postganglionic cell bodies?

a. Acetylcholine
b. Norepinephrine
c. Epinephrine
d. Dopamine
e. Gamma-aminobutyric acid

Q5. Postganglionic parasympathetic neurons secrete acetylcholine at their neuroeffector junctions to act on visceral tissues. What specific class of receptors receives this transmitter on the effector organs?

a. Nicotinic ligand-gated ion channels
b. Muscarinic acetylcholine receptors
c. Alpha-1 adrenergic receptors
d. Beta-1 adrenergic G-protein coupled receptors
e. Mechanically gated sodium channels

Q6. While most sympathetic postganglionic fibers are adrenergic, exceptions exist. Which of the following effector organs is innervated by sympathetic postganglionic fibers that release acetylcholine to act on muscarinic receptors?

a. The smooth muscle lining the coronary arterioles
b. The sinoatrial node pacemaking cells of the myocardium
c. Most systemic sweat glands (and some blood vessels in skeletal muscle)
d. The pilomotor smooth muscles attached to hair follicles
e. The cortical cells of the adrenal gland stroma

Q7. The adrenal medulla plays a unique role in enhancing the function of the sympathetic nervous system. Embryologically and functionally, the secretory cells of the adrenal medulla are modified versions of which autonomic component?

a. Postganglionic parasympathetic neurons containing nicotinic loops
b. Preganglionic sympathetic axons originating in the cervical spine
c. Postganglionic sympathetic neurons, which lose their axons and secrete hormones directly into the bloodstream upon excitation by preganglionic fibers
d. Chemoreceptive glomus cells that monitor plasma carbon dioxide
e. Epithelial goblet cells that synthesize thick mucopolysaccharide fractions

Q8. When the sympathetic nervous system triggers a mass discharge, the adrenal medulla secretes its hormones into the circulation. What is the standard proportion of catecholamines released by the human adrenal medulla?

a. 100 percent pure norepinephrine with zero epinephrine
b. 50 percent norepinephrine and 50 percent dopamine
c. 97 percent acetylcholine and 3 percent epinephrine
d. Approximately 80 percent epinephrine and 20 percent norepinephrine
e. Exactly 95 percent alpha-blockers and 5 percent beta-stimulators

Q9. Circulating catecholamines released by the adrenal medulla differ from the norepinephrine released locally at sympathetic nerve terminals in which of the following ways?
a. They are completely destroyed by plasma albumin within two milliseconds.
b. They cannot cross into any peripheral tissue due to their high molecular mass.
c. They have a longer duration of action because they are cleared slowly from the blood, and they can excite cells that lack direct sympathetic nerve innervation.
d. They cause immediate, profound bradycardia via direct muscarinic mimicry.
e. They act exclusively as low-affinity visual pigments within the retina.
Q10. Norepinephrine released by postganglionic sympathetic fibers binds to adrenergic receptors to cause localized vasoconstriction in most vascular beds. Which specific receptor subtype mediates this vasoconstrictive response?
a. Beta-1 adrenergic receptors
b. Beta-2 adrenergic receptors
c. Muscarinic M2 receptors
d. Nicotinic autonomic G-proteins
e. Alpha-1 adrenergic receptors
Q11. A patient with severe asthma is prescribed an inhaler that triggers bronchodilation to reduce airway resistance. Which autonomic receptor subtype must this drug selectively stimulate to relax bronchiolar smooth muscle?
a. Alpha-1 adrenergic receptors
b. Beta-1 adrenergic receptors
c. Muscarinic M3 receptors
d. Beta-2 adrenergic receptors
e. Nicotinic ganglionic channels
Q12. What primary effect does activation of the sympathetic nervous system exert on the sinoatrial node and myocardium to facilitate a fight-or-flight response?
a. Increased heart rate and increased contractility mediated by beta-1 adrenergic receptors
b. Profound hyperpolarization that slows the heart rate to less than ten beats per minute
c. Complete structural inactivation of the voltage-gated calcium channels
d. Switch to anaerobic glycolysis driven by vagal acetylcholine release
e. Localized vasoconstriction that completely occludes the left coronary artery
Q13. Activation of the parasympathetic nervous system dominates during rest and digest periods. Which of the following physiological changes is a classic result of parasympathetic stimulation?
a. Mydriasis (pupillary dilation) and relaxation of the ciliary muscle
b. Global vasoconstriction of all skeletal muscle arterioles
c. Total inhibition of all gastrointestinal secretory glands
d. Pupillary constriction (miosis), increased gastrointestinal motility, and decreased heart rate
e. Acceleration of glycogenolysis inside the hepatic sinusoids
Q14. What occurs to the overall function of the cardiovascular and respiratory systems when the sympathetic nervous system undergoes a coordinated mass discharge?
a. Airway resistance increases while cardiac output drops to near-zero levels.
b. Total peripheral resistance decreases while heart rate slows down significantly.
c. Arterial blood pressure increases, heart rate increases, coronary blood flow increases, and bronchioles dilate to optimize tissue oxygenation.
d. The central chemoreceptors undergo complete structural necrosis within minutes.
e. Autonomic ganglionic transmission switches entirely to utilizing glycine.
Q15. Muscarinic receptors and nicotinic receptors are both cholinergic, but they utilize entirely different transduction mechanisms. What is the structural nature of the nicotinic receptors found on postganglionic somas in autonomic ganglia?
a. G-protein coupled receptors linked to adenylate cyclase enzymes
b. Tyrosine kinase receptors that cross-link upon ligand binding
c. Intracellular nuclear receptors that alter genetic translation profiles

d. Fibrous anchoring networks bound directly to interstitial collagen sheet layers
e. Ligand-gated ion channels that open directly upon acetylcholine binding to allow rapid sodium influx and depolarization

Q16. The concept of "sympathetic tone" and "parasympathetic tone" describes the continuous baseline activity of the autonomic nervous system. What is the primary homeostatic value of this baseline tone?

a. It allows a single autonomic division to both increase and decrease the functional activity of an organ by changing its firing rate above or below the baseline level.
b. It keeps all visceral organs locked in a state of permanent, spastic contraction.
c. It completely isolates the peripheral tissues from any central nervous center commands.
d. It prevents the kidneys from ever performing the process of pressure diuresis.
e. It forces the intrapleural pressure to remain perfectly fixed at atmospheric zero.

Q17. What occurs to an effector organ's sensitivity to circulating catecholamines several weeks after its direct sympathetic nerve innervation has been surgically removed?

a. The organ becomes completely unresponsive to all forms of chemical stimulation.
b. The organ undergoes rapid structural regression and transforms into adipose tissue.
c. The organ exhibits denervation supersensitivity, becoming significantly more responsive to circulating epinephrine and norepinephrine due to an up-regulation of receptors.
d. The resting membrane potential of the tissue cells drops to minus 150 millivolts.
e. The tissue switches exclusively to utilizing acetylcholine for all motor events.

Q18. The integration and ultimate control of autonomic functions are coordinated by superior structures within the central nervous system. Which brain region functions as the primary organizational head for autonomic regulation?

a. The primary somatosensory cortex in the postcentral gyrus
b. The lateral hemispheres of the cerebellar vermis
c. The dorsal horn gray matter laminae of the spinal cord
d. The hypothalamus
e. The compact bone matrix surrounding the cranium

Q19. When an individual experiences a sudden drop in systemic blood pressure, the baroreceptor reflex alters autonomic outflow. What specific change in autonomic discharge is triggered by the medullary centers?

a. Increased parasympathetic output to the heart to induce bradycardia
b. Increased sympathetic discharge accompanied by reciprocal inhibition of parasympathetic activity, leading to vasoconstriction and increased heart rate
c. Total shutdown of all autonomic nerve pathways to allow venous blood pooling
d. Widespread secretion of acetylcholine by the adrenal cortex cells
e. Immediate structural collapse of the terminal bronchioles to conserve pressure

Q20. Which of the following autonomic parameters is correctly paired with its specific physiological effect on the gastrointestinal system?

a. Sympathetic stimulation – Massive activation of mucosal secretory glands
b. Parasympathetic stimulation – Spastic contraction of the internal anal sphincter
c. Sympathetic stimulation – Widespread relaxation of the muscular wall coats
d. Parasympathetic stimulation – Inhibition of peristaltic motility waves
e. Sympathetic stimulation – Complete destruction of the enteric nervous plexuses

Official Answer Key

Questions	Questions	Questions

1-8	9-16	17-20
Q1: d	**Q9:** c	**Q17:** c
Q2: e	**Q10:** e	**Q18:** d
Q3: d	**Q11:** d	**Q19:** b
Q4: a	**Q12:** a	**Q20:** c
Q5: b	**Q13:** d	
Q6: c	**Q14:** c	
Q7: c	**Q15:** e	
Q8: d	**Q16:** a	